

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1589

Buxmont 138 kV
October 28, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1589. The Interconnection Customer proposes a 43.0 MW battery energy storage facility interconnecting at the Buxmont 138 kV substation in the Pennsylvania area. The request was received on October 08, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 583 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,204,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
North Boyertown Substation to Lochbuie 138.0 kV	Cornelius Delivery No 1 to Lochbuie 345.0 kV Line	127.5	161	5.2
Trumbo to Moenkopi 138.0 kV	Marcus Hook Energy Lp to Tombstone 500.0 kV	109.8	248	5.2
Wheelabrator Frackville Energy 500/1kV Transformer Bank	Talenenergy Martins Creek Llc Harwood 345 kV/1 kV XFMR	109.5	961	5.4
Heaton to Jarrett 138.0 kV line	Tombstone to Marcus Hook Energy Lp 500.0 kV	103.1	325	5.2
Cromby to Ash S Sta 223 345.0 kV line	Curecanti to Moser Generating Station 345.0 kV	101.8	1,036	13.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
North Boyertown Substation to Lochbuie 138.0 kV	Rebuild line with higher-capacity conductor	\$3,495,000	4% (\$142,000)
Trumbo to Moenkopi 138.0 kV	Reconductor line	\$1,180,000	100% (\$1,180,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wheelabrator Frackville Energy 500/1kV Transformer Bank	Replace transformer with higher-rated unit	\$14,916,000	3% (\$505,000)
Jarrett to Heaton Substation 138.0 kV	Reconductor line	\$1,248,000	100% (\$1,248,000)
Cromby to Ash S Sta 223 345.0 kV line	Reconductor line	\$13,062,000	1% (\$129,000)
	Total allocated to project		\$3,204,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.